
AI-Driven Smart Home Energy Optimizer

A capability-driven home-energy platform with explainable automation
and bill-saving estimated in Vietnamese đồng

Product walkthrough · Hanoi University of Science and Technology

What you asked for

Four things shaped the product. We built to each.

- **Many device types**, each with its own controls — without a hand-coded screen per device.
- **Automation** a resident *may* set; if they don't, the system should find the habit and suggest the rule.
- A **mock device** stack, since no physical hardware was available.
- *The priority*: let the customer **see the bill they save** from a suggested rule — in real money.

Everything that follows is a working, end-to-end answer to these four points.

The product at a glance

Live monitoring

Per-device power, energy in kWh, top consumers — refreshed within seconds.

Capability-driven control

Controls render themselves from each device's own capability description.

Explainable rules

Readable WHEN-THEN automation with conflict checks and an audit log.

Habit recommendations

Mined from real usage, ranked by saving, accepted only on your say-so.

Bill-saving in VND

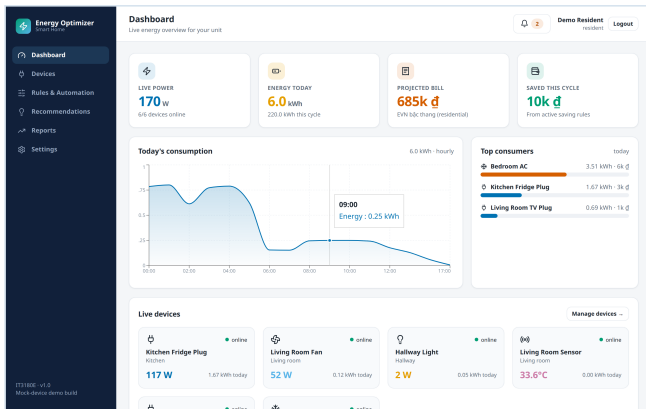
Every rule shows its monthly saving *before* you commit, then tracks it.

Safe by default

Auto-action is opt-in, undoable, never powers off a safety-critical device.

Live monitoring

- Instantaneous power **per device**, refreshed within seconds.
- Energy in **kWh** for today, the billing cycle, and any range.
- **Top consumers** ranked by day, week, month.
- A silent device is marked **offline** after 60 s and dropped from the live total.



Capability-driven control

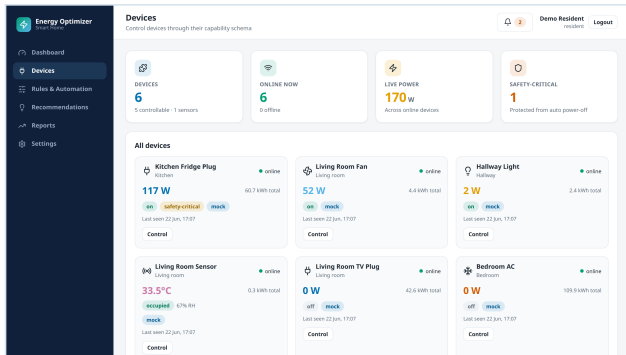
No hard-coded screen per device.

The back-end publishes a **capability description**; the app renders the right control:

- **toggle** on / off
- **range** fan speed, target temperature
- **enum** mode

Each command is **validated** and returns **success**, **rejected**, or **timeout**.

A new device type needs no new control code.



Explainable automation rules

Each rule reads as **WHEN** a condition **THEN** an action, over **time, day, tariff, state or sensor** — with conflict-flagging and an immutable audit log.

The screenshot displays the 'Rules & Automation' section of the 'Energy Optimizer Smart Home' application. The interface includes a dark sidebar with navigation links: Dashboard, Devices, Rules & Automation (selected), Recommendations, Reports, and Settings. The main content area is titled 'Rules & Automation' with a subtitle 'WHEN-THEN rules, conflicts and auto-actions'. In the top right corner, there is a notification bell icon with '2' alerts, a user profile for 'Demo Resident' (resident), and a 'Logout' button. Below the header, there are four summary cards: 'ACTIVE RULES' showing 2 total rules, 'AUTO-ACTIONS' showing 2 actions to be applied automatically, 'EST. MONTHLY SAVING' of 271k đ from active rules, and a '+ New rule' button with the text 'Build a WHEN-THEN rule'. The bottom section lists two specific rules. The first rule, 'Raise Bedroom AC to 26°C at night', is for 'Bedroom AC · Bedroom' and saves 216k đ/mo. Its condition is 'WHEN the time is between 22:00 and 06:00 THEN set Bedroom AC temperature to 26°C'. It is marked as 'enabled' and 'auto' (from recommendation), with checkboxes for 'Enabled' and 'Auto-apply' both checked. The second rule, 'Living Room TV Plug · Living room', saves 55k đ/mo. Its condition is 'WHEN the time is between 18:00 and 23:00 THEN turn off Living Room TV Plug'. It is also marked as 'enabled' and 'auto', with checkboxes for 'Enabled' and 'Auto-apply' both checked. Each rule card includes 'History' and 'Delete' buttons.

Habit recommendations

When no rule exists, the system **mines the habit** from **7+ days** of usage and offers a readable rule with a **rationale** and a ranked **VND saving** — **advisory** until *Accept*.

The screenshot displays the 'Energy Optimizer Smart Home' interface. On the left is a dark sidebar with navigation links: Dashboard, Devices, Rules & Automation, Recommendations (highlighted), Reports, and Settings. The main content area is titled 'Recommendations' with the subtitle 'Explainable saving ideas from your habits'. In the top right corner, there is a notification bell with '2' alerts, a 'Demo Resident resident' profile, and a 'Logout' button. Below the main title, a button 'Analyze my usage' is present. Two recommendation cards are shown:

- #1 Turn off Living Room TV Plug overnight** (suggested)
 - EST. SAVING / MONTH: **13.715 đ**
 - WHEN the time is between 00:00 and 06:00 THEN turn off Living Room TV Plug**
 - Rationale: Living Room TV Plug draws about 0.15 kWh every night (00:00–06:00) while it is likely unused. Switching it off in that window avoids the standby draw.
 - Based on data from 08 Jun, 11:22 – 22 Jun, 11:22
 - Buttons: **✓ Accept**, Dismiss
- #2 Turn off Hallway Light when the room is empty** (suggested)
 - EST. SAVING / MONTH: **4.928 đ**
 - WHEN the room is empty for 15 min THEN turn off Hallway Light**
 - Rationale: Hallway Light stays on for about 0.05 kWh/day while the room is unoccupied. Turning it off after the room is empty for 15 minutes removes that waste.
 - Based on data from 08 Jun, 11:22 – 22 Jun, 11:22
 - Buttons: **✓ Accept**, Dismiss

The priority: bill saving in real money

The customer must **see the money saved** by a suggested rule.

- A 14-day **baseline** per device sets the reference.
- $\text{Saving} = \sum (\text{baseline} - \text{expected}) \times \text{tariff}$ — **transparent**, not a black box.
- The **monthly VND estimate** shows *before* you save a rule or accept a recommendation.
- The **accrued saving** this cycle is tracked, and a rule is re-checked if reality drifts $\pm 20\%$.

Suggested rule

Turn off TV plug at 19:00

Estimated monthly saving

$\approx 38\,000$ đ

shown before you accept

One building, clear roles

The platform models a whole apartment building, with access scoped by role.

Administrator — building owner

Read-only overview of every unit, sets the building tariff, onboards or offboards residents. Never operates a unit's devices, sees no resident's on/off state.

Resident — tenant of one unit

Sees only their own unit. Controls its devices, authors rules, accepts or dismisses recommendations, sees their own savings.

Authentication and role-based access throughout; a resident can never read another unit's data.

Safe and trustworthy by design

The system **advises and estimates**, but acts on devices only with explicit consent.

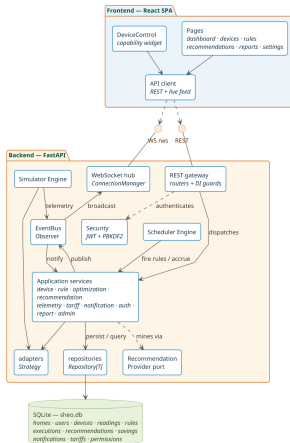
- Automatic action is **off by default** — opt in per rule.
- Every auto-applied action has a **2-minute undo**.
- A safety-critical device is **never auto powered off**.
- Compressor and out-of-range commands are **rate-limited and rejected**.

Deleting a rule keeps its history; nothing happens silently.

Under the hood (briefly)

- **FastAPI** back-end; **React** single-page app over REST + a live WebSocket feed.
- A **capability schema** is the single source of truth for UI rendering *and* command validation.
- A deterministic **mock simulator** stands in for hardware — end-to-end and hardware-ready.
- **67 automated tests** across monitoring, control, rules, savings, safety and access control.

Figure 7 — Component Diagram



Delivered against the brief

- ✓ Controls driven by each device's own capability.
- ✓ Habits mined and suggested as readable rules.
- ✓ Full mock-device stack, hardware-ready.
- ✓ Bill saving shown in **VND**, before you commit.

Ready for your walkthrough

We'll demonstrate any function live.

Thank you